

UniGarment: Topology-Guided Texture Normalization for Simulation-Ready Garment Digitization

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Figure 1: UniGarment recovers sewing-pattern topology and fabric texture from a single photograph, producing GarmentCode assets compatible with industry sewing-pattern tools. Because the output is a sewing pattern rather than a static mesh, the structured result supports texture editing, re-draping on different bodies, and physics-based animation.

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Abstract

Recovering a simulation-ready 3D garment asset from a single in-the-wild photograph remains a long-standing challenge. Prior

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methods address topology estimation and texture normalization in isolation and without closed-loop quality control. We present **UniGarment**, a closed-loop framework that couples the two stages through vision-language verification: a topology stage predicts a GarmentCode sewing pattern, a critic checks panel-boundary alignment against the photograph, the verified pattern localizes which image regions are fabric, and a final critic judges the draped result and re-triggers whichever stage failed. The recovered sewing pattern is what makes the output editable and re-drapeable rather than a static mesh. We evaluate pattern accuracy against genuine ground truth by rendering GarmentCode specifications through the same cloth simulator used at inference, so every evaluation image carries the exact pattern it was produced from. On this 156-garment benchmark the system reaches Panel IoU 0.357 and Seam Accuracy 0.594, and on in-the-wild photographs all garments drape successfully. Our analysis isolates two concrete bottlenecks. Panel count is recovered exactly for only 17% of garments, and seam accuracy degrades from 0.723 to 0.478 as garment complexity grows. It also shows that closed-loop behaviour is governed less by architecture than by how discriminative the critic's score is, a dependency we make explicit.

CCS Concepts

• **Computing methodologies** → **Computer vision; Image synthesis; Shape modeling; Neural networks.**

Keywords

garment digitization, sewing pattern estimation, fabric texture normalization, topology-guided normalization, multimodal language model, diffusion model, panel-guided texture, GarmentCode

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1 Introduction

Garment digitization, i.e., converting a photograph of a garment into a simulation-ready 3D asset, underpins a broad spectrum of multimedia applications: physically plausible virtual try-on, editable garment animation in films and games, personalized digital wardrobes for embodied AI, and AI-assisted fashion design. A simulation-ready asset requires two complementary components: a *sewing pattern* encoding how flat panels are cut and assembled (enabling re-draping on arbitrary bodies), and a *texture map* capturing intrinsic fabric appearance in a tileable, lighting-free form (enabling photorealistic rendering). Neither alone suffices: a mesh without a pattern is non-editable and non-simulatable, while a pattern without texture cannot be rendered. Garment digitization is therefore a *multimodal joint reasoning* problem that fundamentally differs from general-purpose 3D reconstruction, and it shares that character with other multimedia domains whose outputs are structured programs rather than flat signals [3, 14, 63].

Existing methods tackle these two components separately. Sewing-pattern estimators [2, 32, 73] recover GarmentCode [26] panel structures from images but produce no texture. Texture extraction methods [31, 68] recover tileable fabric maps but treat the garment surface as a structureless monolith, ignoring panel boundaries entirely. Broader clothed-human reconstruction [28, 60, 71] and general 3D generation [29, 54, 72] achieve high visual realism but provide no sewing semantics or simulation compatibility. Critically, *all* these methods run open-loop: a mispredicted panel layout silently becomes the spatial prior the texture stage trusts, with no mechanism to detect or revise it, so errors compound across stages.

We argue that bridging these deficiencies hinges on a long-overlooked structural insight: a garment's sewing panels define natural *flat-foldable surface regions* where each panel undergoes an approximately locally consistent warp relative to its canonical flat shape. This means that normalizing texture region by region, conditioned on the recovered pattern as a spatial prior, is a far better-conditioned problem than normalizing the entire surface at once. In other words, *the sewing pattern is not merely the output of the topology stage, but the critical conditioning signal for the texture stage*. We call this principle *topology-guided texture normalization*. Based on this insight, we propose **UniGarment** (Figure 1), a closed-loop framework for garment digitization that explicitly couples structure and appearance with iterative error correction.

Concretely, UniGarment composes two pre-trained predictors with four vision-language critics. A *topology predictor* ($\mathcal{A}_{\text{topo}}$) emits GarmentCode sewing patterns, which a *topology critic* ($\mathcal{V}_{\text{topo}}$) checks against the photograph. The verified pattern states which garment groups the person wears, and a *region critic* ($\mathcal{V}_{\text{tex}}$) uses that to localize one fabric region per group, which a *texture normalizer* ($\mathcal{A}_{\text{tex}}$) turns into a tileable map. A *seam critic* ($\mathcal{V}_{\text{seam}}$) screens the tiles, and a *final critic* judges the draped, body-attached render. Both predictors are used as released, so the contribution lies in how they are coupled and verified rather than in the predictors themselves.

Measuring pattern accuracy requires images paired with ground-truth sewing patterns, which in-the-wild photographs cannot provide. We obtain such pairs directly by rendering GarmentCode specifications through the same cloth simulator used at inference, so each evaluation image carries the exact pattern it came from. On the resulting 156-garment benchmark the system reaches Panel IoU 0.357 and Seam Accuracy 0.594, with the best reconstruction at 0.866. On 181 in-the-wild photographs every garment drapes successfully. The same measurements expose where the approach breaks down: exact panel count is recovered for only 17% of garments, seam accuracy falls from 0.723 on simple garments to 0.478 on complex ones, and the loop engages on 3 of 181 inputs because our stage-level critics' verdicts never reach the retry decision.

Contributions.

- (1) **Closed-loop garment digitization.** A single photograph yields a complete, simulation-ready, *editable* garment, with vision-language verification at every stage and re-prediction of whichever stage the final critic rejects.
- (2) **Topology-guided texture localization.** The verified sewing pattern determines which image regions are fabric and which garment each belongs to, replacing whole-garment texture

extraction with per-region problems. Removing this guidance causes the texture stage to select an unrelated fabric swatch entirely.

(3) **Ground-truth evaluation without paired photographs.**

We construct an image-to-pattern benchmark by rendering GarmentCode specifications through the inference-time simulator, and release the metric implementations, so pattern accuracy can be measured rather than asserted. The same protocol shows how much of a closed loop’s behaviour is decided by its critic’s score distribution.

2 Related Work

2.1 3D Reconstruction and Generation

Methods based on meshes, NeRF, and 3D Gaussian Splatting have achieved impressive visual results in generating 3D garments from images or text. General-purpose image/text-to-3D methods [6, 27, 29, 40, 50, 54, 72], clothed-human reconstruction [16, 28, 60, 71], and garment-specific texture methods [31, 66, 68] all produce visually compelling outputs. A parallel branch lifts video into 3D and 4D [5, 13] or synthesises human-centric video and avatars directly [4, 15, 37, 55, 56], and inherits the same limitation. However, regardless of the underlying representation, their outputs are holistic geometry with *no sewing semantics*: no panel decomposition, no seam connectivity, and no dimensional parameters, making them non-resizable, non-re-drapable, and incompatible with industry-standard simulation engines (CLO3D, Style3D, Optitex). This is analogous to the distinction between 3D meshes and CAD programs in manufacturing: the garment industry relies on *sewing patterns* as the structured program that drives virtual pattern-making, physics simulation, and production output. UniGarment complements these methods by producing *structured assets with sewing semantics*, while resolving their texture quality issues at panel boundaries through topology-guided normalization.

2.2 Structured Garment Generation

Another line of work adopts *sewing patterns* as the core garment representation. Early approaches recover patterns from 3D point clouds or single images [25, 32]; more recent methods adopt the parametric GarmentCode representation [26] and leverage multimodal language models or autoregressive generation [2, 49, 73]. The same move from static geometry to an editable program recurs elsewhere in asset creation, in vector animation, rigging and articulated characters [7, 11, 46, 47]. Their outputs are *editable and simulatable*, directly driving physics-based draping and size grading. However, *none produces texture*, and all operate *open-loop* without verification. Furthermore, the paired data needed to jointly model topology and texture is absent from all existing datasets. UniGarment adopts GarmentCode while bridging both gaps: the predicted pattern determines where fabric is sampled from for texture recovery, and VLM verification closes the prediction loop at both stages.

2.3 Agentic Visual Generation

LLM/VLM-driven agents are moving visual generation toward multi-agent collaboration and closed-loop verification [36, 44, 52]. At the scene level, agentic frameworks have been applied to indoor scenes [18, 39, 42, 61], city-scale generation [21], 4D worlds [70],

and scene crafting with reason-act-reflect paradigms [53, 59]. At the object and image levels, agents decompose generation and self-refine across procedural 3D modelling [34, 45], multimodal 3D creation [10], compositional image planning [57, 62], multi-agent coordination [19, 43], and structured generation of graphics, code, and meshes [1, 17, 48, 51, 58]. A related line makes the agent’s output verifiable by routing it through executable code [8, 9], and benchmarks probe how reliably such agents behave [12, 20, 67]. How reliably such critics read a rendered image depends in turn on the multimodal models behind them and on what their visual tokens retain [22, 33, 35]. The central trend is the *closed-loop verification paradigm* [59, 64, 65, 70]: VLM critics detect structural errors invisible to pixel/token-level losses but apparent from rendered outputs. UniGarment inherits this paradigm but introduces *domain-specific* garment verification (panel-boundary alignment, seam colour continuity, topology validity) and *panel-specific* correction directives, which generic scene critics lack.

3 Method

3.1 Framework Overview

UniGarment recovers a simulation-ready 3D garment asset from a single in-the-wild photograph. As shown in Figure 2, the system composes two pre-trained predictors with four vision-language critics inside one outer refinement loop. This section describes the system exactly as implemented and evaluated; where a component is weaker than the design it instantiates, we say so rather than describing the intent.

Stages. Given an image I , one outer round proceeds as follows.

- (1) $\mathcal{A}_{\text{topo}}$ predicts a GarmentCode sewing pattern P from I , emitted per garment group (upper and lower, or a single whole-body group).
- (2) $\mathcal{V}_{\text{topo}}$ judges the rendered pattern against I .
- (3) The verified pattern determines which garment groups exist; $\mathcal{V}_{\text{tex}}$ then localizes one fabric region per group in I .
- (4) $\mathcal{A}_{\text{tex}}$ normalizes each region into a tileable texture.
- (5) $\mathcal{V}_{\text{seam}}$ inspects the tiles and flags any to re-normalize.
- (6) The pattern is assembled and draped, and a final critic $\mathcal{V}_{\text{final}}$ compares I with the body-attached render, either accepting the result or naming which stage to re-run.

The loop terminates on acceptance or when the iteration budget is exhausted.

Instantiation. Both predictors are used as released, not retrained. $\mathcal{A}_{\text{topo}}$ is the ChatGarment [2] image-reconstruction model, a LLaVA-1.5-7B [30] backbone fine-tuned to emit GarmentCode. $\mathcal{A}_{\text{tex}}$ is the released FabricDiffusion [68] texture model. All four critics are one hosted general-purpose vision-language model, prompted differently per role and reached over an OpenAI-compatible endpoint; every number in this paper was produced with that critic. Assembly and draping use GarmentCodeRC [26] cloth simulation. Consequently the contribution of this work lies in the coupling and verification policy, not in the individual predictors, and we make no claim of improved topology prediction over ChatGarment itself.

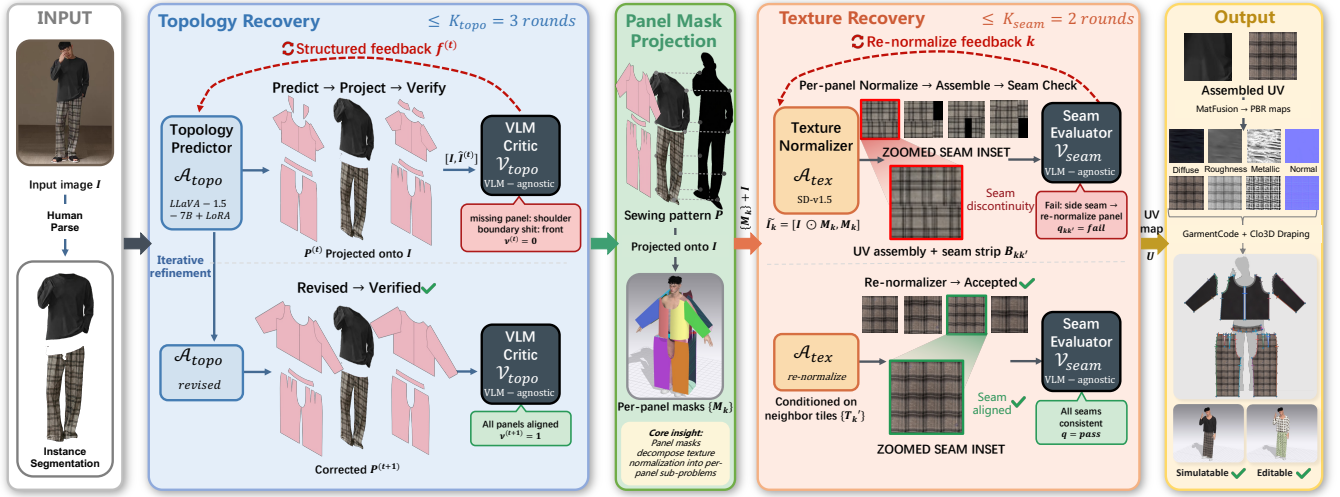


Figure 2: UniGarment framework. The topology stage predicts a GarmentCode sewing pattern, which a VLM critic checks against the photograph. The verified pattern determines the garment groups, and therefore which image regions are sampled as fabric and which garment each texture belongs to. A seam critic screens the generated tiles, and the assembled pattern is draped with GarmentCodeRC cloth simulation before a final critic judges the body-attached render.

3.2 Sewing-Pattern Representation

We use GarmentCode [26] in its GarmentCodeRC form. A pattern P encodes, per panel k , a 2D boundary polygon Q_k given as normalized vertices, together with a set of seam edges S mapping boundary segments of one panel to their partner panel, and scalar parameters covering length, sleeve length, collar depth and ease. This representation is what makes the output editable and re-drapeable: the same P can be resized or re-simulated on a different body, which a reconstructed mesh cannot.

It is also what makes evaluation possible. Because P has the same form as the ground-truth specifications distributed with GarmentCodeData, a prediction can be scored against a reference pattern directly; the supplementary material gives the exact definitions.

3.3 Topology Verification

Motivation. The topology stage predicts in a single forward pass with no downstream quality signal. Mismatched seam edges, wrongly sized panels, or missing components propagate into every later stage: an incorrect pattern determines which garment groups exist, and therefore where fabric is sampled from and how the result is assembled.

Critic. $\mathcal{V}_{\text{topo}}$ receives I together with the rendered pattern and returns a structured judgement: an accept/reject signal, a score, and a natural-language description of where predicted boundaries diverge from visible garment structure.

Implementation limitation. Two things about this critic are weaker than the design it instantiates, and we state both because they bound what Section 4.6 can claim.

First, $\mathcal{V}_{\text{topo}}$'s verdict is *recorded but not acted on*. Re-prediction is triggered by $\mathcal{V}_{\text{final}}$ alone; a topology rejection does not by itself cause the pattern to be re-predicted. On Real-181 the topology critic

rejected 43 of 184 predictions (23%) and none of those rejections started a retry (Table 5).

Second, even when a retry does occur, the critic's feedback is not consumed by the predictor: it is re-invoked without conditioning on $f^{(t)}$, because the released ChatGarment interface accepts no such conditioning. A topology retry is therefore a re-run rather than a correction, and any gain it produces comes from sampling variation, not from the critic's diagnosis. Routing the topology verdict into the retry decision, and conditioning re-prediction on it, are the two concrete fixes this analysis points to; the second additionally requires a predictor exposing that interface.

3.4 Topology-Guided Fabric Localization

Motivation. A garment photograph contains skin, background, accessories and often several garments at once. A texture model applied to the whole image has no way to know which pixels are fabric, which garment they belong to, or how many distinct fabrics are present. The recovered pattern answers exactly these questions: it states how many garment groups the person is wearing and what each one is.

Region selection. The verified pattern supplies the garment groups; for each group, $\mathcal{V}_{\text{tex}}$ localizes one clothing-only fabric region in I , excluding skin and background, and returns it with a confidence and a justification. This is the operative sense in which topology guides texture in our system: guidance is at the granularity of *garment groups*, not per-panel masks. Projecting each panel polygon Q_k to a dense per-panel image mask would give finer conditioning, but requires a texture model that accepts mask conditioning; the released model does not, so we do not claim it.

The effect is nonetheless decisive. Without this guidance the texture stage falls back to a fixed interior heuristic and, on a striped top

worn with dark trousers, selects a single swatch from the trousers and applies it to the whole outfit.

Normalization. $\mathcal{A}_{\text{tex}}$ maps each selected region to a tileable texture. On a retry, the critic’s directive is passed back to region selection so that a different region may be chosen, and the sampling seed is advanced by a fixed stride per attempt so that a retry is not a near-duplicate of the attempt it replaces.

3.5 Seam and Final Verification

$\mathcal{V}_{\text{seam}}$ inspects the generated texture tiles and returns the subset to re-normalize. We note that it examines the tiles themselves rather than an assembled UV atlas, so it detects unusable or inconsistent tiles rather than measuring cross-panel discontinuity directly.

As with $\mathcal{V}_{\text{topo}}$, the returned subset is recorded rather than acted on: the texture stage runs a single pass, so a flagged tile is not re-normalized within the round (Table 5).

$\mathcal{V}_{\text{final}}$ closes the outer loop. It compares I with the body-attached render after assembly and draping, and returns an accept decision, a score, and the set of stages to re-run. A rejection naming texture re-enters region selection with the critic’s directive; one naming topology re-runs prediction and, because a changed pattern invalidates the previous fabric assignment, texture as well.

Acceptance rule. The critic returns both a binary decision and a numeric score, and we treat the score against a configured threshold as authoritative in both directions rather than letting the binary decision stand. What we did not anticipate is that this choice turns out not to matter, because the score itself barely varies: on Real-181 the final critic returned exactly 0.80 on 181 of 184 evaluations, 0.05 twice, and 0.0 once. Any threshold in $(0.05, 0.80]$ therefore produces identical behaviour, and the loop re-engages only when the critic emits an outright failure score. Section 4.6 reports what that leaves.

The general point follows from our own case: a closed loop is only as discriminative as its critic’s score, so a system reporting closed-loop gains should show that score’s distribution, not just its acceptance threshold.

3.6 Assembly and Draping

Each garment group is assembled from its pattern and draped on a body with cloth simulation, producing a simulated mesh, a UV texture and front/back renders. A garment that the simulator refuses to assemble, typically because predicted seams cannot be stitched, counts as a simulation failure, which is what Simulation Success Rate measures.

Groups are currently simulated as independent scenes and their renders composited. This is adequate for scoring an individual garment but does not model interaction between layers, and the composite can misrepresent occlusion where an upper garment should hang over a lower one. Joint multi-garment simulation would remove this limitation.

3.7 Data and Evaluation

Jointly supervising a topology predictor and a texture normalizer would need data pairing sewing patterns with panel-aligned texture observations, absent from existing datasets; that is why this system composes pre-trained predictors rather than training its own.

For *evaluation*, however, pairing an image with a ground-truth sewing pattern is sufficient, and can be obtained directly: we render GarmentCodeData [24] specifications through the same cloth simulator used at inference, so each evaluation image carries the exact pattern it was produced from. This yields the GCD-156 benchmark used throughout Section 4; the supplementary material details its construction.

4 Experiments

Every number reported in this section was measured on the system described in Section 3 using the evaluation code released with this paper. Where a quantity could not be measured, no number is reported.

4.1 Setup

The topology stage predicts GarmentCode sewing patterns from a single image and is instantiated with the ChatGarment image-reconstruction model [2] (LLaVA-1.5-7B backbone [30], released fine-tuned checkpoint). The texture stage is FabricDiffusion [68] conditioned on critic-selected fabric regions. All four critics are served by a single hosted vision-language model behind an OpenAI-compatible endpoint, and the same critic produced every number reported here. Garments are assembled and draped with GarmentCodeRC [26] cloth simulation. All experiments run on a single NVIDIA L20.

Benchmarks. We evaluate on two complementary sets.

GCD-156 measures sewing-pattern accuracy against genuine ground truth. We sample GarmentCodeData [24] specifications and render each one through the same cloth simulator, giving an (image, ground-truth pattern) pair for every garment: the method sees only the render, and its predicted pattern is scored against the specification the render was produced from. Of 172 sampled garments, 156 rendered successfully (91%). We emphasise that these are *synthetic renders, not photographs*; GCD-156 measures pattern accuracy, not robustness to real imagery.

Real-181 is the union of the Dress4D and CLOSE evaluation splits: 181 photographs of real clothed people, used to measure end-to-end reconstruction fidelity where no ground-truth pattern exists. Three baselines run on all 181; the remainder need per-image manual preparation or produce no mesh we can render uniformly, so they are reported on a ten-image subset that every method covers. Both scopes are given, and the sample size is stated in every table rather than elided.

Metrics. **Panel IoU** rasterises each predicted and ground-truth panel polygon into a common normalised frame, matches them with a Hungarian assignment on pairwise IoU, and averages over $\max(|\hat{P}|, |P|)$ panels so that hallucinated and missing panels are both penalised. **Seam Accuracy** is the F1 score over stitched panel pairs after the same matching. We score seams at the panel-pair level rather than requiring identical edge indices, because a predicted panel legitimately carries a different number and ordering of edges than its reference. **SSIM** and **LPIPS** [69] compare the input photograph with the final body-attached render after cropping both to their subject bounding box; this measures end-to-end reconstruction fidelity and is *not* the texture-versus-reference protocol used in prior texture work. **CLIP-Sim** [41] and **DINOv2** [38]

report embedding cosine similarity between the same pair. **SSR** (Simulation Success Rate) is the fraction of garments that produce a non-degenerate simulated mesh.

4.2 Comparison Across Method Families

Table 1 compares every method for which outputs could be produced on the same ten in-the-wild photographs. To keep the comparison about reconstruction rather than about each project’s preview settings, we re-rendered every method’s own output mesh ourselves under one shared camera, lighting and resolution, with UV textures sampled directly.

The raw ordering invites the wrong reading, so three things need stating plainly.

The families do not solve the same problem. Clothed-human reconstruction and generic 3D generation return a dressed body, semantically much closer to the photograph than a garment is, and image metrics reward that directly. PSHuman leading on DINOv2 does not make it better at garment digitization: it produces no sewing pattern, so its output cannot be resized, re-draped or edited, and SSR does not apply to it at all.

SSIM rewards matching empty background. Methods whose subject fills little of an otherwise empty frame score well on SSIM while placing last on the semantic measures, so we read CLIP-Sim and DINOv2 as the informative columns.

Ten images is too few to separate methods. This table establishes which families produce what kind of output, not a ranking, and Sewformer emits 2D patterns with no mesh, hence no image entries. Table 2 repeats the measurements over all 181 photographs, where the ordering differs from the one here.

At 181 images we place last on every image metric, and we report that. HiLo and TripoSG reconstruct the dressed body and BCNet an untextured shell, all rendered by us under one camera. A textured, simulated, panel-decomposed garment is simply further from a photograph of a dressed person, pixel for pixel, than a body-shaped mesh is. We read this as these metrics rewarding silhouette agreement rather than digitization quality, not as our method reconstructing garments worse; what this paper supports quantitatively is pattern accuracy against ground truth (Table 3).

4.3 Qualitative Comparison

Figure 3 compares our results against garment-specific reconstruction (BCNet, HiLo, SIFU, PSHuman), general-purpose 3D generation (TripoSG, TRELLIS.2, Hunyuan3D), surface painting (Paint3D), and sewing-pattern estimation (SewFormer, ChatGarment) on the same inputs.

The distinction that matters here is not visual polish. Several generic 3D generators produce cleaner-looking surfaces, but their output is a single undifferentiated mesh: it cannot be resized, re-draped on a different body, or edited panel by panel, because no sewing semantics were ever recovered. The sewing-pattern rows are the only ones whose output is a garment in the manufacturing sense. We therefore restrict our quantitative evaluation to pattern accuracy, where the comparison is meaningful, rather than reporting image similarity across methods that solve different problems.

4.4 Sewing-Pattern Accuracy

Table 3 reports pattern accuracy on GCD-156, stratified by ground-truth garment complexity.

Across all 156 garments the system reaches Panel IoU 0.357 (median 0.358, $\sigma = 0.170$) and Seam Accuracy 0.594 (median 0.595, $\sigma = 0.223$). The best reconstruction reaches Panel IoU 0.866.

Two failure modes dominate. First, **panel count is rarely exact**: only 17% of predictions match the reference panel count, while 57% over-segment and 26% under-segment. Because Panel IoU averages over the larger of the two panel sets, this miscounting caps the achievable score even when individual panels align well. Second, **seam accuracy degrades monotonically with garment complexity** ($0.723 \rightarrow 0.554 \rightarrow 0.478$), whereas Panel IoU stays roughly flat. Panel *shapes* therefore remain recoverable on complex garments while their *connectivity* does not: the model draws plausible pieces but increasingly mis-assigns which edges join.

4.5 End-to-End Reconstruction

Table 4 reports end-to-end fidelity on Real-181, comparing each input photograph with the final simulated, textured render.

Of the 181 inputs, 180 complete the pipeline and every one of those simulates successfully; the remaining garment fails before simulation, giving 99.4% overall. We report the rate over all attempted inputs rather than over the scored subset, since scoring requires a render that only exists once simulation has already succeeded. The absolute values should be read with the domain gap in mind: the input is a photographed person against a real background, while the render is a textured garment on a grey mannequin over a flat ground. DINOv2 similarity is correspondingly the lowest of the four measures, because it is the most sensitive to exactly that global scene difference. These metrics are useful for *relative* comparison under a fixed rendering setup, and we caution against comparing them across papers with different render conventions.

4.6 Closed-Loop Verification

Table 5 reports how far the closed loop actually engages over the full Real-181 benchmark, measured from the per-round records of all 181 runs.

The loop is largely inert on this benchmark, and we report that rather than a favourable subset. 178 of 181 garments are accepted after a single round. The three that are not were scored 0.05, 0.05 and 0.0 by the final critic, which are outright failures rather than borderline cases, and all three were accepted at 0.80 after one retry. The loop’s measured contribution on Real-181 is thus the recovery of three hard failures, not a general refinement effect.

Two separate causes, both worth naming. The first is the score distribution described in Section 3.5: a critic that answers 0.80 to almost everything cannot drive a loop, whatever threshold it is compared against.

The second is a routing gap in our own implementation, and it is the larger of the two. The two stage-level critics reject at a rate that would have been useful if consulted: $\mathcal{V}_{\text{topo}}$ rejected 43 of 184 predictions and $\mathcal{V}_{\text{seam}}$ flagged two to four panels on 40 of 184. Both verdicts are written to the run manifest and neither is read by the retry decision, which consults $\mathcal{V}_{\text{final}}$ alone. So roughly a

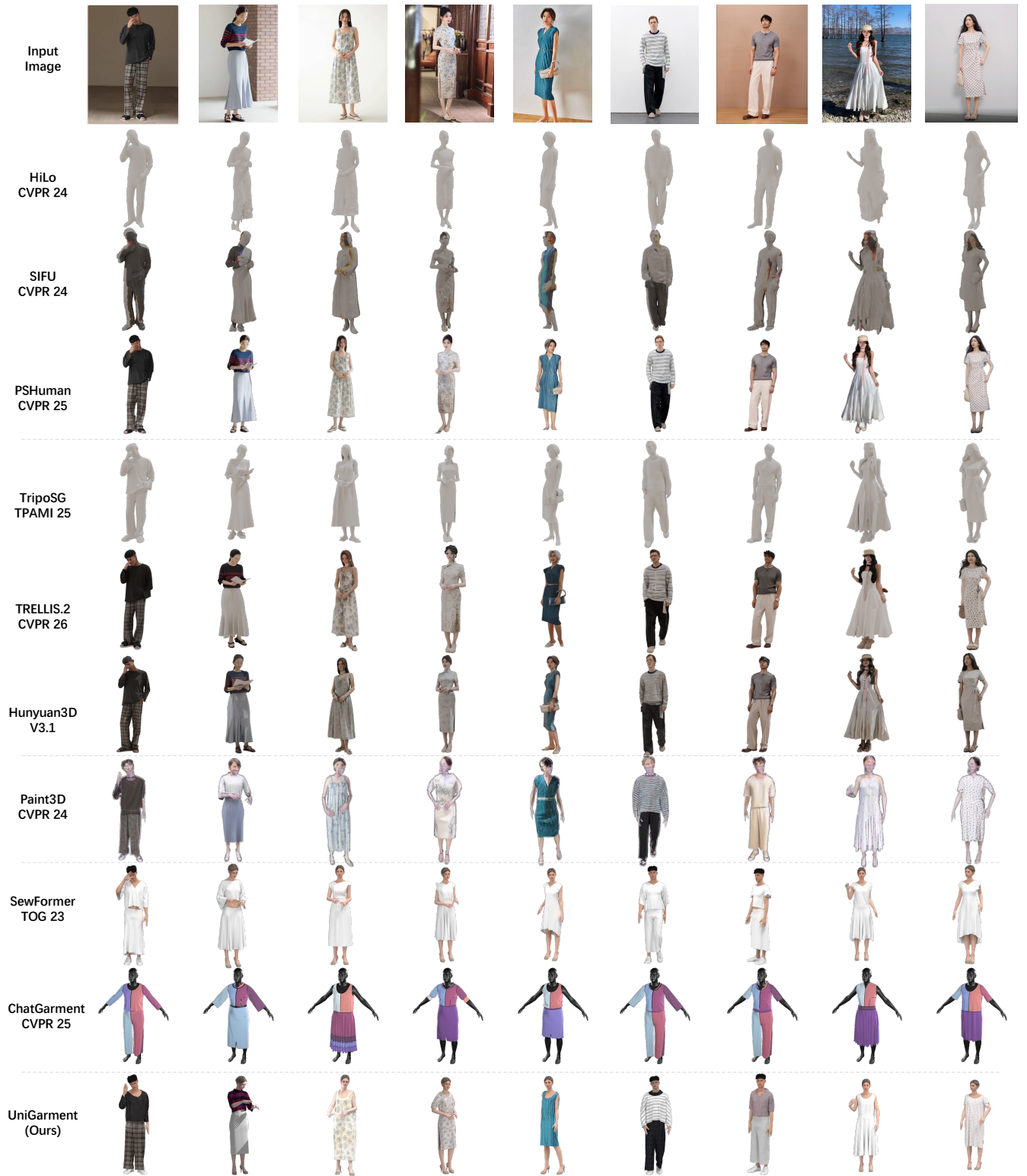


Figure 3: Qualitative comparison on ten in-the-wild photographs against garment-specific and general-purpose reconstruction methods. Only the sewing-pattern methods (SewFormer, ChatGarment, and ours) yield an editable, re-drapeable asset; the remaining rows produce holistic geometry with no panel decomposition.

Table 1: Comparison across method families on ten in-the-wild photographs, every result re-rendered by us from the method’s own output mesh under one shared camera and lighting. Topo = predicts a sewing pattern; Tex = produces texture; Edit = re-drapeable and editable; SSR = Simulation Success Rate, defined only for pattern methods. Best within each family in bold. [†]reconstructs a dressed body, not a garment; [§]textures a supplied mesh and covers 9 of the 10.

Method	Family	Topo	Tex	Edit	SSIM↑	LPIPS↓	CLIP-Sim↑	DINOv2↑	SSR↑
<i>Clothed-human reconstruction</i>									
PSHuman [†] [28]	Human recon	×	✓	×	0.308	0.696	0.733	0.644	–
SIFU [†] [71]	Human recon	×	✓	×	0.365	0.714	0.557	0.210	–
HiLo [†] [60]	Human recon	×	×	×	0.409	0.734	0.610	0.262	–
<i>General-purpose 3D generation</i>									
TRELLIS.2 [54]	Generic 3D	×	✓	×	0.283	0.719	0.661	0.539	–
TripoSG [29]	Generic 3D	×	×	×	0.361	0.736	0.618	0.349	–
<i>Texture generation</i>									
Paint3D [§] [66]	Texture gen	×	✓	×	0.309	0.719	0.617	0.233	–
<i>Garment / sewing-pattern methods</i>									
BCNet [23]	Garment recon	×	×	×	0.458	0.802	0.609	0.183	–
Sewformer [32]	Sewing pattern	✓	×	✓	–	–	–	–	–
UniGarment (Ours)	Sewing pattern	✓	✓	✓	0.472	0.596	0.573	0.266	100.0

Table 2: The same measurements over all 181 real photographs, for the methods we could run at that scale. Ordering differs from the ten-image table, and the two untextured reconstruction baselines lead on every image metric.

Method	Tex	SSIM ↑	LPIPS ↓	CLIP ↑	DINO ↑
HiLo [†] [60]	×	0.574	0.406	0.787	0.545
TripoSG [29]	×	0.479	0.466	0.779	0.534
BCNet [23]	×	0.546	0.535	0.741	0.449
UniGarment (Ours)	✓	0.478	0.571	0.706	0.374

Table 3: Sewing-pattern accuracy on GCD-156, against ground-truth GarmentCode specifications. Stratified by the number of panels in the reference pattern.

Subset	<i>n</i>	Panel IoU ↑	Seam Acc. ↑
Simple (≤ 6 panels)	56	0.312	0.723
Medium (7–12)	57	0.388	0.554
Complex (> 12)	43	0.373	0.478
All	156	0.357	0.594

Table 4: End-to-end reconstruction on 181 real human photographs (Dress4D and CLOSE). Input photograph compared with the final body-attached render.

	<i>n</i>	SSIM ↑	LPIPS ↓	CLIP-Sim ↑	DINOv2 ↑	SSR ↑
Ours	180	0.488	0.572	0.725	0.314	99.4%

Table 5: Closed-loop engagement over all 181 Real-181 runs. Two of the three critics reject at a useful rate, but only the final critic is wired to the retry decision, so only its rejections change anything.

Critic	Rejections	Rate	Retries caused
$\mathcal{V}_{\text{topo}}$	43 / 184	23.4%	0
$\mathcal{V}_{\text{seam}}$	40 / 184	21.7%	0
$\mathcal{V}_{\text{final}}$	3 / 184	1.6%	3
<i>Resulting rounds per run</i>			
One round only	178 / 181	98.3%	
Two rounds	3 / 181	1.7%	
Three rounds	0 / 181	0.0%	

fifth of predictions and a fifth of texture sets were flagged as wrong by the very critics built to catch them, and none was redone: our stage-level verification is, as shipped, instrumentation rather than control. This is invisible end-to-end, since Table 4 reads the same whether those 83 rejections were acted on or discarded.

5 Conclusion

We presented **UniGarment**, a framework coupling sewing topology and fabric texture through vision-language verification, and measured pattern accuracy against ground truth by rendering GarmentCode specifications through the inference-time simulator. It reaches Panel IoU 0.357 and Seam Accuracy 0.594 on 156 such garments, and every in-the-wild garment drapes. The same measurements mark the road ahead: exact panel count is recovered for only 17% of garments, seam accuracy falls from 0.723 to 0.478 with complexity, and a fifth of our critics’ rejections never reach the retry decision.

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